Name:	Class:	Date given:
		Date due in:

Acids and **Alkalis**

1. What does the pH of a substance tell you? [1]

2. Litmus paper changes colour to show whether a substance is an acid or an alkali. Describe how universal indicator is different to litmus paper. [2]

3. Match the substances to their pH by drawing a line between the substance and the pH scale. [4]



water	washing up liquid	hydrochloric acid	ammonia

4. Use the words from the box to complete the sentences. [4]

water	neutralisation	neutral	salt

a. When an acid and alkali react together, the solution formed is _____.

b. This is called a _____ reaction.

c. The products of this type of reaction are always a _____ and _____.

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5. Below are the steps of a method for carrying out a neutralisation reaction and making a salt. Rewrite the steps into the correct order. [6]

   - Measure 25cm^3 of sodium hydroxide solution.
   - Stop adding acid when the solution tests neutral.
   - Use a pipette and add hydrochloric acid drop by drop.
   - Leave the rest to evaporate overnight and form crystals.
   - After every few drops, remove a small sample to test the pH.
   - Pour the solution into an evaporating dish and heat until only a small amount of liquid remains.

   Step 1: _____

   Step 2: _____

   Step 3: _____

   Step 4: _____

   Step 5: _____

   Step 6: _____

6. From method above:

   a. Name the alkali. [1] _____

   b. Name the acid. [1] _____

   c. Name the salt. [1] _____

Learning Outcomes (tick if achieved)

Q1,2	I can describe pH and indicators	
Q3	I can identify common substances on the pH scale	
Q4	I can describe a neutralisation reaction	
Q5	I can describe the method for making a salt	
Q6	I can identify reactants and products in a neutralisation reaction	

Acids and Alkalis **Answers**

1. What does the pH of a substance tell you? [1]

   Acidity, or how acidic or alkali a substance is.

2. Litmus paper changes colour to show whether a substance is an acid or an alkali. Describe how universal indicator is different to litmus paper. [2]

   Range of dyes / spectrum of pH identifiable

   Identifies weak / strong acids and alkali / neutral

3. Match the substances to their pH by drawing a line between the substance and the pH scale. [4]

   water - 7 **washing up liquid - 9** **hydrochloric acid - 1** **ammonia - 13**

   Allow +/- 1 pH on the scale for each answer.

4. Use the words from the box to complete the sentences. [4]

   a. When an acid and alkali react together, the solution formed is **neutral**.

   b. This is called a **neutralisation** reaction.

   c. The products of this type of reaction are always a **salt** and **water**.

5. Below are the steps of a method for carrying out a neutralisation reaction and making a salt. Rewrite the steps into the correct order. [6]

   Step 1: **Measure 25cm^3 of sodium hydroxide solution.**

   Step 2: **Use a pipette and add hydrochloric acid drop by drop.**

   Step 3: **After every few drops, remove a small sample to test the pH.**

   Step 4: **Stop adding acid when the solution tests neutral.**

   Step 5: **Pour the solution into an evaporating dish and heat until only a small amount of liquid remains.**

   Step 6: **Leave the rest to evaporate overnight and form crystals.**

6. From the method above:

   a. Name the alkali [1]: **sodium hydroxide**

   b. Name the acid [1]: **hydrochloric acid**

   c. Name the salt [1]: **sodium chloride**